

Q1

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Step 1: Basic concepts

- GCV (Gross calorific value): Total heat released when fuel is completely burnt and product are cooled so water vapor condenses

- NCV (Net calorific value): Heat released when product are not cooled, so water remains as vapour

- Step 2: Combustion of Hydrogen

Hydrogen in fuel burns as
$$2H_2 + O_2 \rightarrow 2H_2O$$

- 2kg hydrogen produces 18kg of water
⇒ 1kg of hydrogen produces 9kg of water

- water formed from hydrogen in fuel

if fuel contains H% hydrogen, then:

- Hydrogen per kg fuel = $\frac{H}{100}$ kg

- water formed = $9 \times \frac{H}{100} = 0.09H$ kg

- More hydrogen $\rightarrow$ more water vapour - more heat loss $\rightarrow$ larger difference

Q3

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Feature	Bomb calorimeter	Boy's calorimeter
System type	Closed system	Open system
Fuel type	Solid and liquid fuels	Gaseous Fuels
Pressure	High pressure	Atmospheric pressure
Combustion	Complete the combustion in sealed bombs	Continuous combustion
Measurement	measure GCV directly	measure NCV directly
Water condensation	Condensed	Remains as vapour

Q4

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Boy's calorimeter is used to determine the calorific value of LPG.

Reason:-

- LPG is gaseous fuel
- Boy's Calorimeter is determined for continuous combustion of gaseous
- It measures NCV directly, which is more practical for gaseous fuels used in real applications

- Step 6:

$$NCV = GCV - 0.09H \times 2$$

where:

$$: H = 5$$

$$: L = 587 \text{ cal/g}$$

$$\text{Heat loss} = 264.15 \text{ cal/g}$$

- Step 7:

$$NCV = 7057 - 264.15$$

$$= 6793 \text{ cal/g}$$

Q2.

→

$$\bullet w = 1.1g$$

$$\bullet \text{volume} = 50 \text{ ml}$$

$$\bullet \text{Normality} = 0.1 \text{ N}$$

$$\bullet \text{Back titration} = 18 \text{ ml of } 0.1 \text{ N}$$

- Step 1

$$\text{Acid consumed} = 50 - 18 = 32 \text{ ml}$$

- Step 2

$$V \cdot N = \frac{100 \times N \times V}{w}$$

 w

where

$$\bullet N = 0.10$$

$$\bullet V = 32 \text{ ml}$$

$$\bullet w = 1.1g$$

04

→ Given -

- ① From 1.00g coal → 0.040g BaSO_4
- ② From 0.60g coal → 1.82g CO_2 , 0.36g H_2O

③ From 1.10g coal

• Acid = 50ml of 0.1N

- Step 1:

Relation:

$$\text{for } \text{BaSO}_4 = \frac{32}{233}$$

$$\text{mass of S} = 0.040 \times \frac{32}{233} = 0.00549\text{g}$$

$$7.8 = \frac{0.00549}{1.00} \times 100$$

$$7.8 = 0.549\%$$

- Step 2

$$\text{C in } \text{CO}_2 = \frac{12}{44}$$

$$\text{mass of C} = 1.82 \times \frac{12}{44} = 0.36\text{g}$$

$$\% \text{C} = \frac{0.36}{0.6} \times 100 = 60\%$$

Q5]

→ Step 1: Corrected temperature θ_{150}
 $\Delta T_{corrected} = 4.2 + 0.08 = 4.28^\circ\text{C}$

Step 2:

$$\text{Water to water equivalent} = 1000 + 200 = 1200\text{g}$$

$$Q = 1200 \times 4.28 = 5136 \text{ Cal}$$

Step 3:

$$Q_{\text{fuel}} = 5136 - 49 = 5087$$

Step 4: Gross calorific value (GCV)

$$\text{GCV} = \frac{5087}{0.35} = 5985$$

Step 5: $\text{NCV} = \text{GCV} - (0.09 \times 11 \times 587)$

$$= 5985 - (0.09 \times 6 \times 587)$$

$$= 5985 - 317$$

$$= 5668 \text{ Cal/g}$$

Q6

→ ① Percentage of sulphur

$$\text{S in } \text{BaSO}_4 = \frac{32}{233}$$

$$\text{mass of S} = 0.030 \times \frac{32}{233} = 0.00412 \text{ kg}$$

$$\% \text{ S} = \frac{0.00412}{0.95} \times 100$$

Q7

→ Given:

- Gas volume = 0.09 m^3
- water circulated = 28.5 kg
- Steam condensed = 0.08 kg
- Temp = 8.6°C

Step 1 :-

$$Q_1 = m \times \Delta T = 28.5 \times 9.6 = 245.1 \text{ kcal}$$

Step 2 -

$$Q_2 = 0.08 \times 587 = 46.96 \text{ kcal}$$

Step 3 :-

$$\text{GCV} = \frac{Q_1 + Q_2}{\text{Gas volume}} = \frac{245.1 + 46.96}{0.09} = 3245 \text{ kcal/m}^3$$

Step-4

$$\text{NCV} = 2723 \text{ kcal/m}^3$$